TRACKING · EFFORTS AND MOST DEMANDING SCENARIOS

PEAK DEMANDS

ATHLETE	DATE	TEST	MSS
Player 9 (anonymised)	2026-10-31	Full match	9.20 m/s

01 SUMMARY

9.66 km
TOTAL DISTANCE
103 m/min · 94 min

646 m
HIGH SPEED
> 5.5 m/s · 28 efforts

7
SPRINTS
> 7.0 m/s · max 9.03 m/s

21 / 25
ACCELERATIONS / DECELERATIONS
> 2.5 m/s²

186 m/min
PEAK 1 MIN
start 68:56

READING

9,661 m in 94 min (103 m/min), of which 646 m above 5.5 m/s and **7 sprints**.

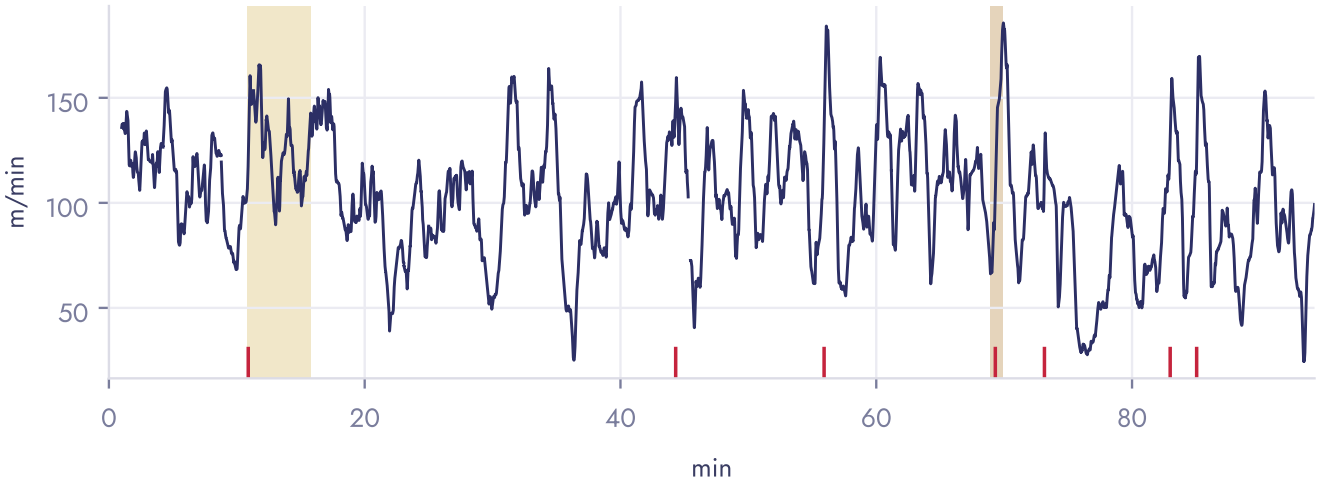
The most demanding minute reaches **186 m/min** (at 68:56), 1.8 times the match average; the next window drops to 77 m/min. Fixed epochs would have underestimated it by 2 %.

Peak intensity falls with duration along a power law of exponent -0.27 (R^2 0.92): a reference for programming drills of different lengths.

5 sequences of repeated high-speed efforts (≤ 60 s recovery), the longest with 2 efforts; 0 meet Spencer's stricter criterion (≥ 3, mean recovery < 21 s).

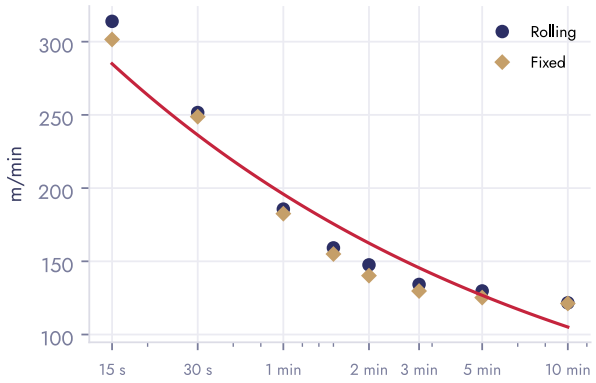
Direction changes: **53 %** of the total acceleration comes from turning rather than from speed changes — a part that classic GPS metrics ignore.

02 THE MATCH MINUTE BY MINUTE



Distance per minute (60-s rolling average). Bands: most demanding 1- and 5-min windows; red ticks: sprints.

03 ▶ MOST DEMANDING SCENARIOS



Peak of the rolling average (dots) and of fixed epochs (diamonds) for each window length; line: fitted power law. $I = 592 \cdot t^{-0.270}$, R^2 0.921.

READING

Over 15 s the peak is 3.0 times the match average; over 10 min, 1.2 times.

Fixed epochs underestimate the peak by 0.3 to 5.0 % depending on duration.

A single peak is unstable from match to match: look at the next window and at how often near-peak passages recur too.

WINDOW	M/MIN	HSR	ACC+DEC	FIXED	START	NEXT WINDOW	≥ 90 %
15 s	314	211	40.0	−3.9 %	69:13	73	3
30 s	252	106	30.0	−1.1 %	68:58	118	2
1 min	186	53	17.0	−1.7 %	68:56	77	4
1 min 30	159	35	12.0	−2.6 %	10:49	107	8
2 min	148	27	14.0	−5.0 %	61:29	103	6
3 min	134	31	10.3	−3.3 %	10:44	130	12
5 min	130	20	7.6	−3.5 %	10:46	111	8
10 min	122	13	7.0	−0.3 %	7:16	90	5

m/min: distance; HSR: m/min above the threshold; acc+dec: efforts per minute; Fixed: underestimation by fixed epochs; Start: time of the peak (min:s); ≥ 90 %: distinct passages at 90 % of the peak.

04 ▸ EFFORTS

TYPE	COUNT
Sprints	7
> 5.5 m/s	28
Accelerations	21
Decelerations	25
Direction changes	91
Repeated sequences	5
Spencer	0
> 85 % MSS	4
> 90 % MSS	2
> 95 % MSS	0

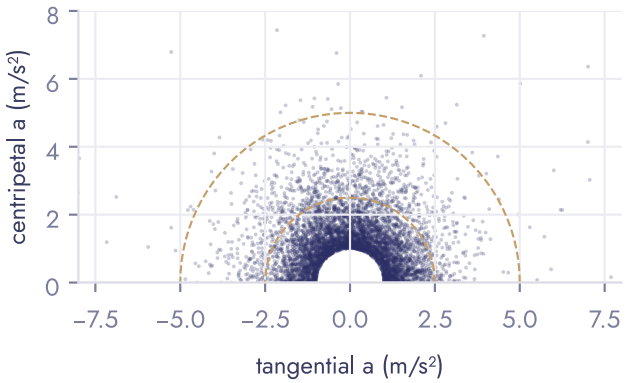
4.1 Repeated-sprint sequences

START	COUNT	RECOVERY	DISTANCE (M)
7:47	2	8 s	21
44:02	2	14 s	28
49:16	2	13 s	36
55:14	2	37 s	46
69:06	2	8 s	52

4.2 Sprints

START	DURATION	MAX SPEED	ENTRY SPEED	M
73:09	3.0 s	9.03	7.02	24
85:03	5.3 s	8.63	7.06	43
69:19	3.0 s	8.18	7.01	23
55:55	1.3 s	7.35	7.02	10
44:19	1.0 s	7.26	7.00	7
10:54	1.7 s	7.23	7.00	12
82:59	2.4 s	7.20	7.00	17

05 ▸ DIRECTION AND EXPOSURE (GPS 3.0)



Each dot is an instant with acceleration ≥ 1 m/s²: speed-change component (horizontal) and direction-change component (vertical).

53 %

DIRECTIONAL SHARE

a_c^2 / $|a|^2$, $|a| \geq 1$ m/s²

5.1 Intensity exposure time (IET)

REFERENCE	≥ 60 %	≥ 70 %	≥ 80 %
Peak 1 min (186 m/min)	36.1 min	17.7 min	5.5 min
Bundle S1min	0.0 min	0.0 min	0.0 min

Estimated Maximal S1min = 28.2 km/h (470 m/min). Estimated Maximal MP1min (Eq. 2): 8.2 W/kg — the paper's reference for a mechanical power not reproduced here.

06 METHOD

Positions smoothed and differentiated with a Savitzky-Golay filter (0.6-s window); tracking jumps removed. The complete acceleration is split into a tangential (speed change) and a centripetal (turning) component; $|a|^2 = a_t^2 + a_c^2$.

Efforts: passages above a threshold held for at least 1 s (speed) or 0.5 s (acceleration). Repeated sequences: at least 2 efforts no more than 60 s apart (Buchheit et al. 2010); Spencer et al. (2004) criterion: at least 3 efforts, mean recovery < 21 s.

Most demanding scenarios: peak of the rolling average for each duration, compared with fixed epochs (Varley et al. 2012; Fereday et al. 2020), power law $I = c \cdot t^n$ (Delaney et al. 2018). The peak alone is unstable (Novak et al. 2021): it is given with its time, the next window and the number of passages at 90 % of the peak.

GPS 3.0 (Buchheit et al. 2026): intensity exposure time (IET) = minutes during which the 60-s rolling average exceeds 60, 70 or 80 % of a reference — the match peak minute and, when MAS and MSS are known, Bundle's maximal 1-min speed (MAS + 0.695·(MSS − MAS)). The paper's mechanical work comes from a proprietary analyser that is not published: it is not reproduced here.

Data: Metrica Sports sample data, game 2, Home team, Player9.

- Buchheit M., Lopez Sagarra A., Boskovic A., Komino P., Norman D. & Hader K. (2026). GPS 3.0: from distance into zones toward better proxies of internal neuromuscular load in elite football. Sport Perf Sci Rep 280.
- Buchheit M. & Simpson B.M. (2017). Player-tracking technology: half-full or half-empty glass? Int J Sports Physiol Perform 12 (Suppl 2), S235–S241.
- Bundle M.W., Hoyt R.W. & Weyand P.G. (2003). High-speed running performance: a new approach to assessment and prediction. J Appl Physiol 95, 1955–1962.
- Buchheit M., Mendez-Villanueva A., Simpson B.M. & Bourdon P.C. (2010). Repeated-sprint sequences during youth soccer matches. Int J Sports Med 31, 709–716.
- Buchheit M., Settembre M., Hader K. & McHugh D. (2023). Exposures to near-to-maximal speed running bouts during different turnarounds in elite football. Biol Sport 40, 1057–1067.
- Spencer M. et al. (2004). Time-motion analysis of elite field hockey, with special reference to repeated-sprint activity. J Sports Sci 22, 843–850.
- Harper D.J., Carling C. & Kiely J. (2019). High-intensity acceleration and deceleration demands in elite team sports competitive match play. Sports Med 49, 1923–1947.
- Varley M.C., Elias G.P. & Aughey R.J. (2012). Current match-analysis techniques' underestimation of intense periods of high-velocity running. Int J Sports Physiol Perform 7, 183–185.
- Fereday K. et al. (2020). A comparison of rolling averages versus discrete time epochs for assessing the worst-case scenario locomotor demands of professional soccer match-play. J Sci Med Sport 23, 764–769.
- Delaney J.A., Thornton H.R., Rowell A.E. et al. (2018). Modelling the decrement in running intensity within professional soccer players. Sci Med Football 2, 86–92.
- Novak A.R., Impellizzeri F.M., Trivedi A., Coutts A.J. & McCall A. (2021). Analysis of the worst-case scenarios in an elite football team. J Sports Sci 39, 1850–1859.
- Whitehead S. et al. (2018). The use of microtechnology to quantify the peak match demands of the football codes: a systematic review. Sports Med 48, 2549–2575.

Report generated by peakdemands 0.1.0. Thresholds and windows are parameters: compare only at equal method.